

Ahmed Mohamed Ashraf Helmy

Backend Engineer — Software Engineer

New Cairo, Egypt

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Professional Summary

Backend Engineer with hands-on experience building scalable backend systems, RESTful and gRPC APIs, and database-driven applications. Skilled in Go (Golang) and Node.js for developing secure, high-performance services in fintech and data-driven environments. Experienced in event-driven architectures, caching, authentication and authorization, and backend system design, with a strong foundation in full-stack development using Angular, React, MongoDB, and SQL.

Education

Master of Computer Science

Oct 2025 – Present

Future University in Egypt (FUE)

Faculty of Computers and Information Technology

Major: Computer Science

Bachelor of Computers and Information Technology

2020 – 2024

Future University in Egypt (FUE), Cairo, Egypt

(In collaboration with the University of Cincinnati, USA)

- Major: Computer Science
- GPA: 3.05

Professional Experience

Backend Engineer — Egyptian Bureau for Engineering (EBE)

Sep 2025 – Present

- Designed, developed, and maintained scalable backend systems supporting secure, multi-region client operations.
- Built RESTful and gRPC APIs using Go (Golang) with clean architecture and strong separation of concerns.
- Applied microservices and distributed systems principles to design reliable backend services.
- Implemented event-driven architecture using RabbitMQ to improve system reliability and service decoupling.
- Integrated Redis caching to optimize performance and reduce response time in high-throughput systems.
- Designed and optimized relational databases (MySQL, SQLite) ensuring efficient queries and data consistency.
- Performed API testing, debugging, and validation using Postman and Insomnia.
- Collaborated with cross-functional teams to deliver scalable, secure, and maintainable backend solutions.

Full Stack Developer Intern — Information Technology Institute (ITI), Egypt **Mar – Aug 2025**

- Designed and developed full-stack applications using Node.js, Express.js, Angular, and React with focus on backend engineering.
- Built RESTful APIs implementing CRUD operations, routing, validation, and database integration using MongoDB.
- Implemented authentication and authorization systems using secure backend logic and JWT.
- Applied OOP and MVC design patterns to build scalable and maintainable applications.
- Designed and managed SQL and NoSQL databases supporting data-driven features.
- Used Docker for containerization and environment consistency during development.
- Developed responsive UI using HTML5, CSS3, and Bootstrap integrated with backend services.
- Gained exposure to Generative AI concepts and modern software engineering practices.

Mobile Application Developer Intern (Flutter) — Next Academy, Egypt **Aug – Sep 2023**

- Developed cross-platform mobile applications using Flutter and Dart.
- Integrated Firebase for authentication, data storage, and real-time features.
- Applied mobile architecture principles, state management, and UI component design.

Certifications

- Artificial Intelligence and Machine Learning Diploma — Route IT Training Center
- Machine Learning — NTI
- Database Fundamentals — Mahara Tech
- Machine Learning — NVIDIA
- Mastering Java Programming — YAT Solutions

Skills

- **Languages:** Go, JavaScript, TypeScript, Java, Python, PHP, C++, C
- **Backend:** Go (Golang), Node.js, Express.js, REST APIs, gRPC, Microservices, Distributed Systems, Event-Driven Architecture, RabbitMQ, Redis, Authentication, Authorization, JWT
- **Frontend:** Angular, React, HTML5, CSS3, Bootstrap
- **Databases:** MySQL, PostgreSQL, SQLite, MongoDB, Firebase
- **Concepts:** OOP, MVC, System Design, API Development, Database Design, Scalable Systems, Performance Optimization
- **Tools:** Git, GitHub, Docker, Postman, Insomnia, Linux
- **AI:** Machine Learning Fundamentals, Generative AI Basics

Selected Projects

Duat – AI-Powered Tourism Platform

- Developed backend-integrated features and AI-driven workflows to support personalized tourism recommendations.
- Integrated APIs and data pipelines between services and machine learning models.
- Technologies: Flutter, Python, TensorFlow, Firebase, Google APIs

Fintech Backend Services Platform

- Built backend services and APIs using Go, REST, and gRPC for financial operations.
- Implemented event-driven workflows using RabbitMQ for reliable message-based processing.
- Integrated Redis caching to improve performance and scalability.
- Designed relational databases for transaction processing and data consistency.
- Technologies: Go, REST APIs, gRPC, RabbitMQ, Redis, MySQL, PostgreSQL

AI Expense Tracker

- Developed backend APIs for expense tracking and analytics workflows.
- Integrated AI-based data analysis for financial insights.
- Technologies: Angular, Firebase, Python

Full-Stack Online Bookstore

- Built backend APIs using Node.js and Express.js including authentication and authorization using JWT.
- Designed database schemas and implemented backend business logic for secure e-commerce workflows.
- Integrated payment processing functionality.
- Technologies: Angular, Node.js, Express.js, MongoDB, JWT, Stripe

Gazar Foods – Cloud Kitchen Platform

- Developed RESTful backend APIs to manage orders, menus, and user data.
- Implemented backend business logic and validation using Node.js and MongoDB.
- Built frontend using Angular to consume APIs and deliver responsive user interfaces.
- Technologies: Angular, Node.js, Express.js, MongoDB

Languages

Arabic (Native), English (Fluent)